

JEE(Main + Advanced) : ENTHUSIAST COURSE [SCORE-I]
ANSWER KEY
PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	B	C	A	C	D	C	C	C
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	B	D	C	C	A	A	A	C	B	A
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	1500	2	17	3	7	75	30	9	5	80

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	A	D	B	D	B	B	D	A	C
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	A	D	B	A	D	A	D	A	D	B
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	4	2	5	35	5	64	6	0	500	72

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	D	B	B	D	C	B	C	B	B
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	D	B	C	C	B	B	B	B	B	C
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	171	728	15	3	1	1	2	1	2	25

HINT – SHEET
PART-1 : PHYSICS
SECTION-I

 1. **Ans (C)**

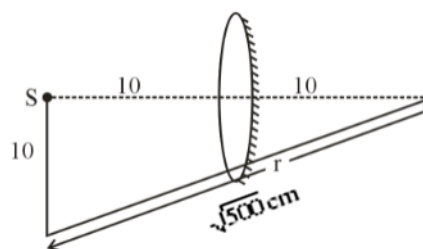
$$r = 10\sqrt{5} \text{ cm}$$

$$I = \frac{P}{4\pi r^2} = \frac{1}{2} \epsilon_0 E_0^2 \times c = \epsilon_0 E_{\text{rms}}^2 \times c$$

$$E_{\text{rms}} = \sqrt{\frac{P}{4\pi \epsilon_0 \times C \times r^2}}$$

$$= \sqrt{\frac{10 \times 9 \times 10^9}{3 \times 10^8 \times 500}} \times 10^{-4}$$

$$= \sqrt{\frac{9}{15}} \times 10^2 \text{ N/C}$$


 2. **Ans (B)**

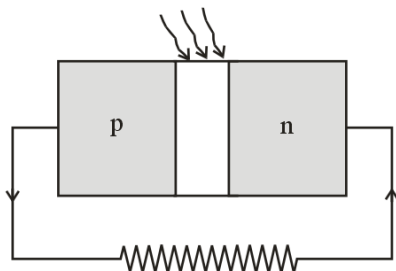
Amplifier in repeater will restore its amplitude and compensate for loss.

loss = -20 dB

so amplification = + 20 dB

3. **Ans (B)**

Circuit diagram of solar cell.



4. **Ans (C)**

Significant figure in value of error must be same

$$m = (5.7890 \pm 0.04) \times 10^{-5} \text{ kg}$$

$$= (5.79 \pm 0.04) \times 10^{-5}$$

5. **Ans (A)**

$$a_1 \cos 60^\circ = a_2 \cos 30^\circ$$

$$a_1 = \sqrt{3} a_2$$

$$N \cos 30^\circ = ma_2$$

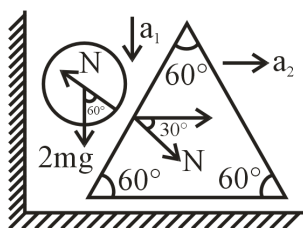
$$2mg - N \cos 60^\circ = 2ma_1$$

$$2mg - \frac{ma_2}{\sqrt{3}} = 2ma_1$$

$$2g = \frac{a_2}{\sqrt{3}} + 2 \times \sqrt{3} a_2 = \frac{7a_2}{\sqrt{3}}$$

$$a_2 = \frac{2\sqrt{3}}{7}g; a_1 = \frac{6}{7}g$$

$$\therefore a_{\text{com}} = \left(\frac{2}{7}\hat{i} - \frac{2\sqrt{3}}{7}\hat{j} \right) \text{ gm/s}^2$$



6. **Ans (C)**

For maximum power through device $\rightarrow$

$$\text{Resistance of device} = R = 45 + 5 = 50 \Omega$$

For rise in temperature,

$$\Delta T \rightarrow 50 = 40 (1 + 0.005 \Delta T)$$

$$1.25 - 1 = 0.005 \Delta T$$

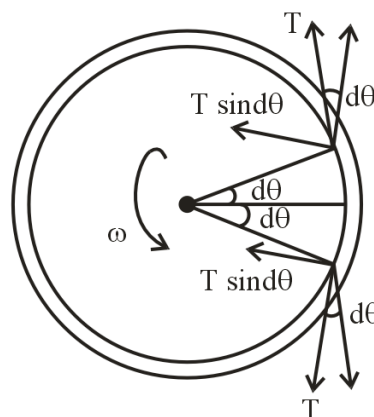
$$\Delta T = \frac{0.25}{0.005} = \frac{250}{5} = 50 \text{ K}$$

$$\text{Temperature} = T_0 + \Delta T = 273 + 50 = 323 \text{ K}$$

7. **Ans (D)**

At any time 't' and angular velocity ' ω ' $\rightarrow$

Required centripetal force will be provided by tension in string



$$2T \sin d\theta = dm\omega^2 R$$

$$dm = \frac{M}{2\pi R} (R \times 2d\theta)$$

$$2T \sin d\theta = \frac{M}{\pi} d\theta + \omega^2 R$$

$$\sin d\theta \approx d\theta$$

$$\therefore 2T d\theta = \frac{M\omega^2 R}{\pi} d\theta; T = \frac{M\omega^2 R}{2\pi}$$

$$\text{For } \omega \rightarrow \alpha = \frac{d\omega}{dt} = \alpha_0 t$$

$$\Rightarrow \int_{\omega_0}^{\omega} d\omega = \alpha_0 \int_0^t t dt \Rightarrow \omega = \left(\omega_0 + \frac{\alpha_0 t^2}{2} \right)$$

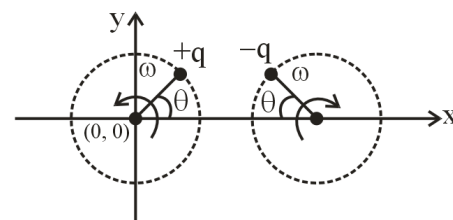
$$\therefore \text{Stress} = \frac{T}{A} = \frac{MR}{2\pi A} \left(\omega_0 + \frac{\alpha_0 t_0^2}{2} \right)^2$$

8. **Ans (C)**

Dipole moment at any instant is given as $\rightarrow$

$$p = q[4R - 2R \cos \theta]$$

$$p = 2qR[2 - \cos \theta]$$



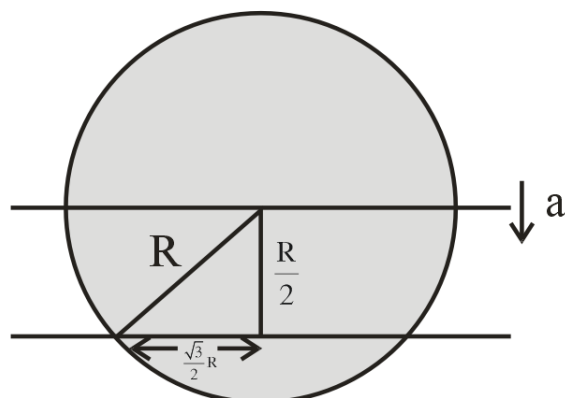
Rate of change of dipole moment

$$= \frac{dp}{dt} = 2qR \frac{d \cos \omega t}{dt}$$

$$\frac{dp}{dt} = 2qR\omega \sin \omega t, \text{ which will be maximum for}$$

$$\omega t = \frac{\pi}{2}; t = \frac{\pi}{2\omega} \text{ sec}$$

9. Ans (C)



Motional emf = $BV\ell$

$$v = \sqrt{2as} = \sqrt{2a \times \frac{R}{2}} = \sqrt{aR}$$

$$\ell = 2 \times \sqrt{R^2 - \left(\frac{R}{2}\right)^2} = 2 \times \frac{\sqrt{3}}{2} R = \sqrt{3}R$$

$$e = BV\ell = B\sqrt{aR} \times \sqrt{3}R = \sqrt{3a}BR\sqrt{R}$$

$$e = \sqrt{3a}BR^{\frac{3}{2}}$$

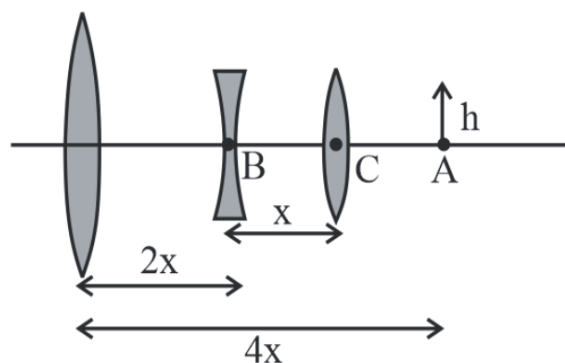
10. Ans (C)

Force = mean pressure $\times$ area

$$= \rho g \frac{H}{2} \times \frac{5}{4} H \times b$$

$$= \frac{5}{8} \rho g H^2 b$$

11. Ans (B)



For Lens 'B' A will be object

$$\frac{1}{v} - \frac{1}{v} = \frac{1}{f}; \frac{1}{v_B} - \frac{1}{2x} = \frac{1}{-x}$$

$$\frac{1}{v_B} = \frac{1}{-x} = \frac{1}{2x}; v_B = -2x$$

Magnification of lens B

$$= -\left(\frac{v_B}{u_B}\right) = -\left(\frac{-2x}{-2x}\right) = -1; M_B = -1$$

For lens 'C' image of lens 'B' will be object

$$\frac{1}{v_C} - \frac{1}{u_C} = \frac{1}{f_C}; \frac{1}{v_C} - \frac{1}{-3x} = \frac{1}{2x}$$

$$\frac{1}{v_C} = \frac{1}{2x} - \frac{1}{3x}; v_C = 6x$$

Magnification of lens C

$$= -\left(\frac{v_C}{u_C}\right) = -\left(\frac{6x}{-3x}\right) = M_C = 2$$

Net magnification of lens B and lens C

$$= (M_B)(M_C) = \frac{h_i}{h_o}$$

$$h_i = -2h, |h_i| = 2h$$

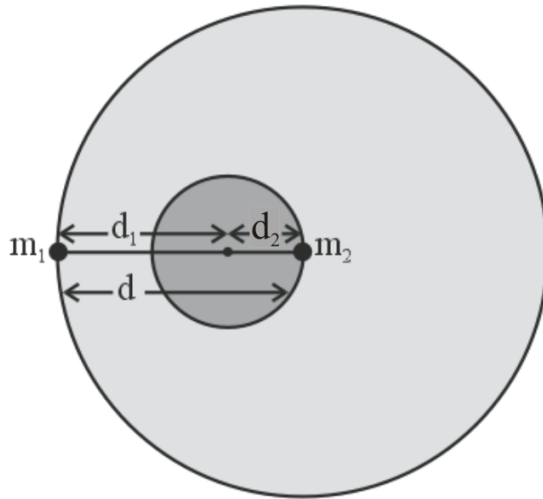
12. Ans (D)

$$\frac{Gm_1m_2}{d^2} = m_1d_1 \frac{4\pi^2}{T^2} \dots (1)$$

$$\frac{Gm_1m_2}{d^2} = m_2d_2 \frac{4\pi^2}{T^2} \dots (2)$$

Adding (1) & (2)

$$\frac{G}{d^2} M = (d) \frac{4\pi^2}{T^2}; \left(\frac{GMT^2}{4\pi}\right)^{\frac{1}{3}}$$



13. Ans (C)

Given each scratch is one half oscillation

$$\Rightarrow 'Q' \text{ amount of heat is produced in time } \frac{\pi}{\omega}$$

$$\therefore \frac{Q}{\left(\frac{\pi}{\omega}\right)} = \frac{KA(T_1 - T_2)}{\ell}$$

$$\Rightarrow T_2 = T_1 - \frac{Q\omega\ell}{\pi KA}$$

14. **Ans (C)**

From figure (A), angular velocity of box should be same as angular frequency of light which is performing S.H.M.

Figure (B), if light now spots box at $\frac{\sqrt{3}H}{2}$

$\Rightarrow$ phase angle covered by light is 60° .

$\Rightarrow \omega t = 60^\circ \dots (i)$

From initial position of box, in same time 't', for it to be spotted by light, its angular displacement will be 30° .

$\Rightarrow \omega't = 30^\circ \dots (ii)$

$$\frac{1}{2} \Rightarrow \frac{\omega}{\omega'} = \frac{60}{30}$$

$$\omega' = \frac{\omega}{2}$$

Adjustment to be done is $\frac{\omega}{2}$

15. **Ans (A)**

A $\rightarrow$ B

$$dQ = dU + d\omega$$

$$0 = E_1 + 0.1 \times 10^6 (0.3 - 0.003)$$

$$E_1 = -29.7 \text{ kJ}$$

B $\rightarrow$ C

$$dQ = dU + d\omega$$

$$-(105 \times 10^3) = E_2 + 0.1 \times 10^6 [0.06 - 0.3]$$

$$E_2 = -110.7 \text{ kJ}$$

$$\frac{E_2}{E_1} = \frac{110.7}{29.7} = 3.72$$

16. **Ans (A)**

$$T d\theta - F_e = (dm)\omega^2 r$$

$$\Rightarrow T d\theta - F_e + F_m - T' d\theta = dm\omega^2 r$$

$$\Rightarrow F_m = T' d\theta$$

$$\Rightarrow \frac{QB d\theta \omega R}{2\pi} = \frac{Y \Delta R d\theta A}{RN}$$

$$\Rightarrow B = \frac{Y A 2\pi}{NQR\omega}$$

17. **Ans (A)**

$$KE = \frac{hc}{\lambda} - \phi$$

$$\lambda_D = \frac{h}{\sqrt{2mK}} = \frac{h}{\sqrt{2m\left(\frac{hc}{\lambda} - \phi\right)}}$$

$$\frac{hc}{\lambda} - \phi = \frac{h^2}{2m\lambda_D^2}$$

$$\frac{h^2}{2m} = A, \quad hc = B$$

18. **Ans (C)**

$$F_1 = m(g + a) = 10(10 + 2) = 120 \text{ N}$$

$$T = m(g + a) = 10(10 + 2) = 120 \text{ N}$$

$$F_2 = 2T \sin 37^\circ$$

$$= 2(120) \left(\frac{3}{5}\right) = 144 \text{ N}$$

$$\frac{F_1}{F_2} = \frac{120}{144} = \frac{10}{12} = 5 : 6$$

19. **Ans (B)**

Angular acceleration remains same w.r.t. all reference frame.

20. **Ans (A)**

$$-\frac{du}{dr} = F = u \left[-12 \frac{R_0^{12}}{r^{13}} + \frac{12R_0^6}{r^7} \right] = 0$$

$$u_0 \left(\frac{12R_0^6}{r^7} \right) \left(-\frac{R_0^6}{r^6} + 1 \right) = 0$$

$r = R_0$ mean position.

Now displaced by 'x' from mean.

$$F = u_0 \left[-12 \frac{R_0^{12}}{(R_0 + x)^{13}} + \frac{12R_0^6}{(R_0 + x)^7} \right]$$

$$= u_0 \left[-12 \frac{R_0^{12}}{R_0^{13}} \left(1 + \frac{x}{R_0} \right)^{-13} + \frac{12R_0^6}{R_0^7} \left(1 + \frac{x}{R_0} \right)^{-7} \right]$$

$$= u_0 \left[-\frac{12}{R_0} \left(1 - \frac{13x}{R_0} \right) + \frac{12}{R_0} \left(1 - \frac{7x}{R_0} \right) \right]$$

$$F = \frac{u_0}{R_0^2} (12 \times 13 - 12 \times 7) x = \frac{72u_0 x}{R_0^2}$$

$$\Rightarrow \omega^2 = \frac{72u_0}{mR_0^2}$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (1500)**

With both ends closed.

$$f_1 = 1680 = \frac{v}{2\ell_1} \Rightarrow \ell_1 = 10 \text{ cm}$$

With both ends open

$$f_2 = \frac{v}{2\ell_2} \Rightarrow \ell_2 = (10 + 0.6 \times 2) \text{ cm} = 11.2 \text{ cm}$$

$$\Rightarrow F_2 = 1500 \text{ Hz}$$

2. Ans (2)

$$[Y] = [V^a A^b F^c]$$

$$[Y] = [M^1 L^{-1} T^{-2}]$$

$$[V^a A^b F^c] = [L^1 T^{-1}] [L^1 T^{-2}]^b [M^1 L^1 T^{-2}]^c$$

$$= [L^1 T^{-a}] [L^b T^{-2b}] [M^c L^c T^{-2c}]$$

$$= [M^1 L^{-1} T^{-2}] = [M^c T^{a+b+c} L^{-a-2b-2c}]$$

$$c = 1 \dots\dots (1)$$

$$a + b + c = -1 \dots\dots (2)$$

$$-a - 2b - 2c = -2 \dots\dots (3)$$

$$a + b + 1 = -1 \Rightarrow a + b = -2, -2b + b = -2$$

$$-a - 2b - 2 = -2 \Rightarrow -a - 2b = 0, -b = -2$$

$$a = -2b, b = 2, a + b + c$$

$$= -4 + 2 + 1 = -1, a = -2 \times 2 = -4$$

3. Ans (17)

Incident ray $\vec{AO}$

$$= -\hat{i} - 2\hat{j} - 3\hat{k}$$

Using vector form of reflection,

$$\vec{r} = \hat{i} - 2(\hat{i}, \hat{n})\hat{n}$$

$$\hat{r} = \frac{-\hat{i} + 2\hat{j} - 3\hat{k}}{\sqrt{14}}$$

$$\Rightarrow \vec{r} = \vec{OB} = -\hat{i} + 2\hat{j} - 3\hat{k}$$

As reflected is completely polarised, refracted and reflected are 11ar

$$\Rightarrow (2\hat{i} + N\hat{j} + 5\hat{k}) \cdot (-\hat{i} + 2\hat{j} - 3\hat{k}) = 0$$

$$\Rightarrow -2 + 2N - 15 = 0$$

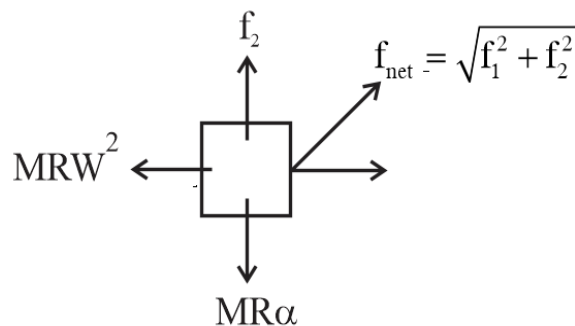
$$\Rightarrow 2N = 17$$

4. Ans (3)

Given at time $t = \frac{1}{\sqrt{2}}$ block just begins to slide.

$\Rightarrow$ Net friction at this instant

Top view F.B.d. : F_{S_L}



$$\Rightarrow f_{net} = \sqrt{(MRw^2)^2 + (MR\alpha)^2}$$

$$\Rightarrow f_{S_L} = \sqrt{[MR(0 + \alpha t)^2]^2 + (MR\alpha)^2}$$

$$\Rightarrow (\mu mg)^2 = (MR\alpha^2 t^2)^2 + (MR\alpha)^2$$

$$\Rightarrow \mu^2 g^2 = R^2 \alpha^4 t^4 + R^2 \alpha^2$$

$$\Rightarrow \mu^2 g^2 = R^2 \alpha^4 \left(\frac{1}{\alpha^2}\right) + R^2 \alpha^2$$

$$\Rightarrow \mu^2 g^2 = 2R^2 \alpha^2 \Rightarrow \alpha = \frac{\mu g}{\sqrt{2}R}$$

$$\therefore W_{friction} = f_2 \cos 0^\circ + f_1 \cos 90^\circ$$

$$= (MR\alpha)(R\theta) = MR \left(\frac{\mu g}{\sqrt{2}R}\right) \left(R \left(\frac{0+1}{2} \alpha t^2\right)\right)$$

$$= \frac{\mu mg}{\sqrt{2}} \left(R \times \frac{1}{2} \alpha \times \frac{1}{\alpha}\right) \Rightarrow W_{friction} = \frac{\mu MgR}{2\sqrt{2}}$$

5. Ans (7)

$$\alpha + \beta = 7$$

$$v = \sqrt{\frac{\gamma RT}{M}} = \sqrt{\frac{3}{5} \frac{RT}{M}}$$

$$\frac{dx}{dt} = \sqrt{\frac{\gamma RT}{M}}; T = T_0 + \left(\frac{T_2 - T_1}{L}\right)x$$

$$dx = \sqrt{\frac{\gamma R}{M} \left(T_0 + \left(\frac{T_2 - T_1}{L}\right)x\right)} dt$$

$$t = \int_0^L \frac{dx}{\sqrt{\frac{\gamma R}{M} \left(T_0 + \frac{(T_2 - T_1)x}{L}\right)}}$$

$$= \frac{2L}{\sqrt{T_1} + \sqrt{T_2}} \sqrt{\frac{3M}{5R}}$$

$$\alpha = 2 \quad \beta = 5$$

6. **Ans (75)**

Intensity after passing first sheet

$$I_1 = 100 \times \cos^2 60$$

$$= 100 \times \frac{1}{4} = 25 \text{ Watt/m}^2$$

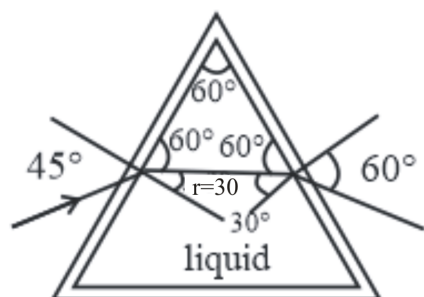
Intensity after passing second sheet

$$I_2 = 25 \cos^2 30$$

$$= 25 \times \left(\frac{\sqrt{3}}{2} \right)^2 = \frac{75}{4}$$

7. **Ans (30)**

Parallel glass slab will not



Produce any derivation.

$$1. \sin 45^\circ = \sqrt{2} \sin r \frac{1}{\sqrt{2}} = \sqrt{2} \sin r; \sin r = \frac{1}{2}$$

$$r = 30^\circ, \delta = (i - r) + (e - r)$$

$$= (i + e) + (e - r)$$

$$= 45 + 45^\circ - 2 \times 30^\circ, \delta = 30^\circ$$

8. **Ans (9)**

$$\lambda = \frac{h}{MV}, V = \sqrt{\frac{\mu RT}{M}}$$

$$\lambda = \frac{h\sqrt{M}}{M\sqrt{\mu RT}}, \lambda = \frac{h}{\sqrt{\mu RMT}}$$

$$\frac{\lambda_{\text{He}}}{\lambda_{\text{H}_2}} = \sqrt{\frac{\mu_{\text{N}_2} M_{\text{x}_2} T_{\text{N}_2}}{W_{\text{He}} M_{\text{He}} T_{\text{He}}}}$$

$$= \sqrt{\frac{7/5 \times 28 \times 400}{5/3 \times 4 \times 300}}$$

$$= \sqrt{\frac{21}{25} \times \frac{28}{4} \times \frac{4}{3}} = \sqrt{\frac{7 \times 28}{25}} = \frac{14}{5} = 2.8$$

9. **Ans (5)**

When charge is maximum, current will be zero.

W_B = energy on capacitor

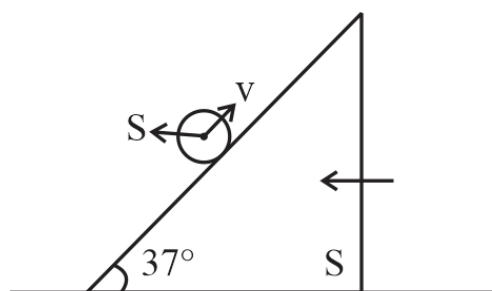
$$q \times E = \frac{q^2}{2C} \Rightarrow q_{\text{max}} = 2CE$$

Current when charge is half of this maximum value

$$CE \times E - \frac{(CE)^2}{2C} = \frac{1}{2} LI^2 \Rightarrow I = \sqrt{\frac{C}{L}} E$$

10. **Ans (80)**

v is velocity of ring wrt wedge.



$$5 = \sqrt{v^2 + 5^2 + 2 \times v \times 5 \times \cos(180 - 37^\circ)}$$

$$\Rightarrow v = 8 \text{ m/s}$$

$$\text{Angular velocity} = \frac{8}{0.1} = 80 \text{ rad/s}$$

PART-2 : CHEMISTRY

SECTION-I

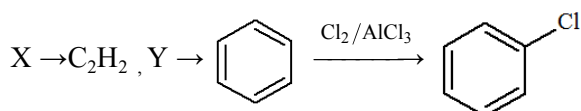
1. **Ans (A)**

Classical smog contains SO_2 , smoke and fog

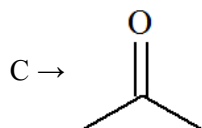
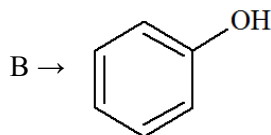
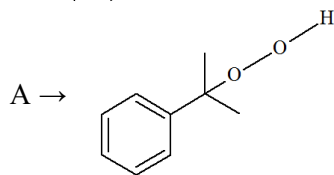
2. **Ans (A)**

Isoelectronic species contain same number of valence e^- , same number of atoms and same shape.

3. **Ans (D)**



4. **Ans (B)**



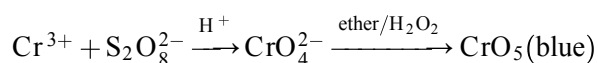
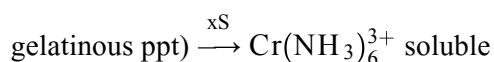
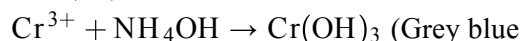
5. **Ans (D)**

$$\Rightarrow \frac{13.6}{2^2} + \text{KE} = 13.6 \times 3^2 \left[\frac{1}{2^2} - \frac{1}{4^2} \right]$$

$$\Rightarrow \text{KE} = 13.6 \frac{3^2}{2^2} \times \frac{3}{4} - \frac{13.6}{4}$$

$$= \frac{13.6}{4} \times \left[\frac{23}{4} \right] = 19.55 \text{ eV}$$

6. **Ans (B)**



7. **Ans (B)**

$$\rho = \frac{z \times m}{a^3 \times N_a} \Rightarrow 1 = \frac{4 \times 30}{a^3 \times N_a}$$

$$\rho' = \frac{4 \times [30 + 50 \times 2]}{a^3 \times N_a}$$

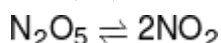
$$\Rightarrow \frac{\rho}{\rho'} = \frac{1}{\rho'} = \frac{30}{130}$$

$$\text{or } \rho' = \frac{13}{3} = 4.33 \text{ g/cc}$$

8. **Ans (D)**

Copper metal forms in reverberatory furnace

9. **Ans (A)**



10. **Ans (C)**

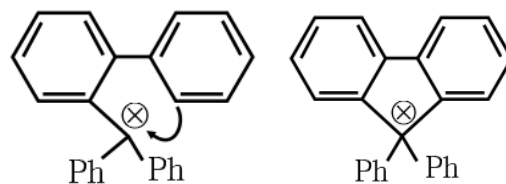
I → S

II → R

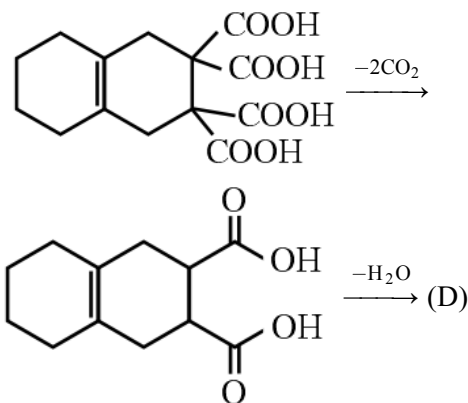
III → S

IV → R

11. **Ans (A)**



12. **Ans (D)**

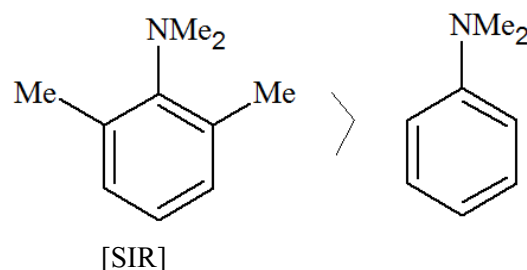


13. **Ans (B)**

$$[\text{H}^+] = \frac{6 \times 10^{-7}}{5 \times 10^{-2} \times 10^{-3}} = 1.2 \times 10^{-2} \text{ mol/L}$$

$$\therefore \text{Time taken} = \frac{1.2 \times 10^{-2}}{6 \times 10^5} = 2 \times 10^{-8} \text{ sec.}$$

14. **Ans (A)**



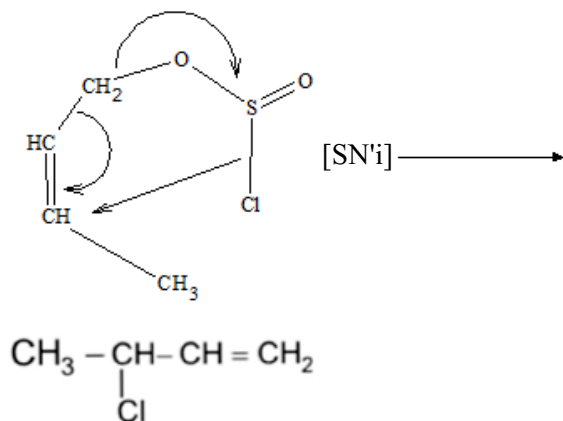
16. **Ans (A)**

CaZ , MgZ are formed and Na^+ goes into solution.

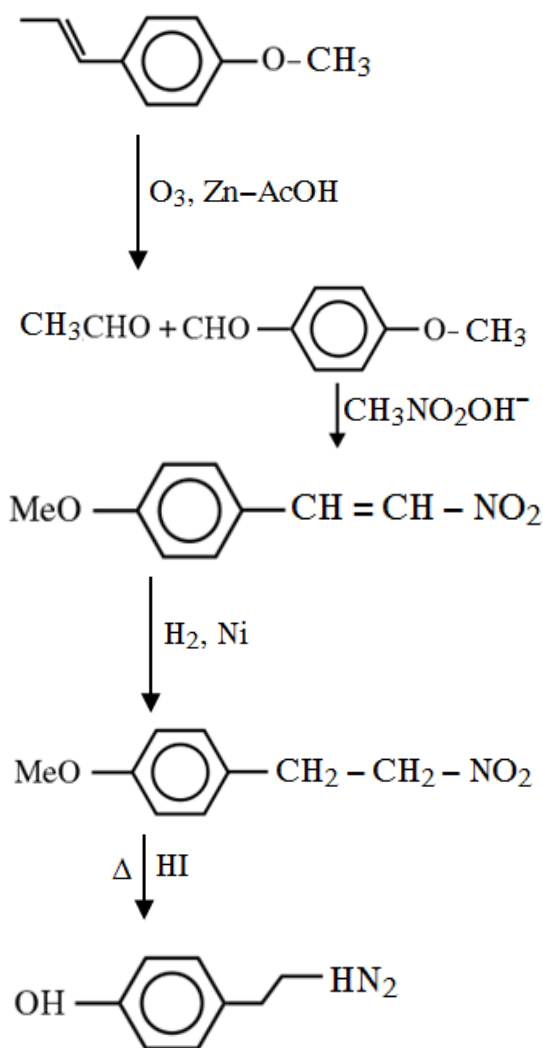
17. **Ans (D)**

ICl_3 is orange solid. Rest options are colourless gases.

18. Ans (A)



19. Ans (D)



20. Ans (B)

(A) Process of precipitating colloidal solution is called coagulation. Hence false.

(B) For colloidal solutions concentration is very small due to very large molar mass and hence their colligative properties are very small as compared to true solutions

$\therefore \Delta T_f$ is lesser for colloidal solution. Hence true.

(C) At CMC surfactant form micelles. Hence true

(D) Micelles and macromolecular colloids are two different types of colloids. Hence false.

PART-2 : CHEMISTRY

SECTION-II

1. Ans (4)

$$s = [\text{C}_6\text{H}_5\text{COO}^-]_0 = [\text{C}_6\text{H}_5\text{COO}^-]$$

$$+ [\text{C}_6\text{H}_5\text{COOH}] = [\text{Ag}^+]$$

$$= [\text{C}_6\text{H}_5\text{COO}^-] \left[1 + \frac{[\text{H}^+]}{K_a} \right]$$

$$K_{sp} = [\text{C}_6\text{H}_5\text{COO}^-] [\text{Ag}^+]$$

$$= \frac{s^2}{1 + \frac{[\text{H}^+]}{K_a}}$$

$$\Rightarrow s = \sqrt{K_{sp} \left\{ 1 + \frac{[\text{H}^+]}{K_a} \right\}}$$

$$\text{In Pure water, } s_2 = \sqrt{K_{sp}}$$

$$\therefore \text{Ratio : } \frac{s}{s_2} = \sqrt{1 + \frac{[\text{H}^+]}{K_a}}$$

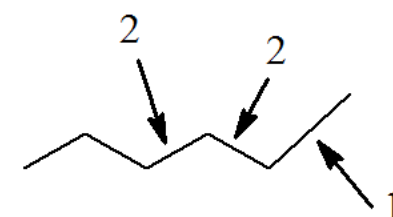
$$= \sqrt{1 + \frac{10^{-3.1}}{6.4 \times 10^{-5}}} = 3.67$$

2. Ans (2)

Only and are soluble. and

are soluble.

3. Ans (5)



4. **Ans (35)**

$$n = \frac{P \cdot V}{R \cdot T};$$

$$n = \frac{26}{760} \times \frac{0.80 \times 3.14 \times 1^2 \times 10 \times 10^3}{0.0821 \times 300}$$

$$= 34.89$$

5. **Ans (5)**

Siderite, Fool's gold, Limonite, Magnetite, Haematite

6. **Ans (64)**

For a 'n' th order reaction,

$$t_{\frac{1}{2}} \propto \frac{1}{a^{n-1}} \text{ where } a = \text{initial conc.}$$

$$\left(\frac{5 \times 10^{-3}}{25 \times 10^{-4}} \right)^{n-1} = \frac{8}{1} \Rightarrow n = 4$$

$$\frac{t_{1/2}}{1} = \left(\frac{5 \times 10^{-3}}{1.25 \times 10^{-3}} \right)^{n-1}$$

$$t_{1/2} = 64 \text{ hr}$$

7. **Ans (6)**

$$\text{Eq. of Cu} = \frac{i \cdot t}{96500} \text{ or } \frac{1.98}{63.6/2} = \frac{i \cdot t}{96500}$$

$$\therefore i \cdot t = Q = 6008.49 \text{ coulomb} = 6 \text{ kC}$$

8. **Ans (0)**

For a given aqueous solution

$$\Delta T_b = K_b \times \text{molality}$$

$$\Delta T_f = K_f \times \text{molality}$$

$$\therefore \frac{\Delta T_b}{\Delta T_f} = \frac{K_b}{K_f} = \frac{RT_b^2}{1000\ell_v} \times \frac{1000\ell_f}{RT_f^2}$$

$$\frac{\Delta T_b}{\Delta T_f} = \frac{T_b^2 \times \ell_f}{T_f^2 \times \ell_v}$$

$$\therefore T_b = 100 + 273 = 373 \text{ K}$$

$$T_f = 0 + 273 = 273 \text{ K}$$

$$\ell_f = 80 \text{ cal g}^{-1};$$

$$\ell_v = 540 \text{ cal g}^{-1}$$

$$\therefore \frac{0.1}{\Delta T_f} = \frac{373 \times 373 \times 80}{273 \times 273 \times 540}$$

$$\therefore \Delta T_f = 0.362$$

$$\therefore T_f = 0.0 - 0.362 = -0.362^\circ\text{C}$$

$$|T_f| = 0.362^\circ\text{C}$$

$$1000 T = 362$$

9. **Ans (500)**

$$\text{Degree of dissociation } (\alpha) = \frac{\Lambda_m}{\Lambda_m^0}$$

$$\Rightarrow \Lambda_m = \Lambda_m^0 \times \alpha$$

$$= \Lambda_m^0 \times \sqrt{\frac{K_a}{C}} \text{ (by Ostwald dilution law)}$$

$$= 390 \times \sqrt{\frac{1.8 \times 10^{-5}}{0.02}}$$

$$= 11.7 \text{ ohm}^{-1} \text{cm}^2 \text{mol}^{-1}$$

$$\text{Conductivity} = \frac{\text{molarity} \times \Lambda_m}{1000}$$

$$= 2.34 \times 10^{-4} \text{ ohm}^{-1} \text{cm}^{-1}$$

$$\text{Also, conductivity} = \text{cell constant} \times \frac{1}{\text{Resistance}}$$

$$\Rightarrow \text{Resistance} = \frac{\text{cell constant}}{\text{conductivity}}$$

$$= \frac{0.117}{2.34 \times 10^{-4}} = 500 \Omega$$

10. **Ans (72)**

$$\left(P + \frac{a}{V_m^2} \right) (V_m) = RT$$

$$PV_m + \frac{a}{V_m} = RT$$

$$PV_m^2 - RTV_m + a = 0$$

$$PV_m^2 - 24V_m + 2 = 0$$

For real gas this equation must have only one root

$$D = 0 \Rightarrow b^2 - 4ac = 0$$

$$(24)^2 - 4 \times P \times 2 = 0$$

$$P = 72 \text{ atm}$$

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (B)**

$$x^2 - ax = 3ay + 2$$

$$\left(x - \frac{a}{2}\right)^2 = 3ay + 2 + \frac{a^2}{4}$$

$$\left(x - \frac{a}{2}\right)^2 = 3a \left(y + \frac{a}{12} + \frac{2}{3a}\right)$$

$$\Rightarrow \text{Let vertex is } (h, k) = \left(\frac{a}{2}, \frac{-a}{12} - \frac{2}{3a}\right)$$

$$h = \frac{a}{2}, K = \frac{-a}{12} - \frac{2}{3a}$$

$$\Rightarrow k = \frac{-h}{6} - \frac{1}{3h}$$

$$\Rightarrow xy = \frac{-x^2}{6} - \frac{1}{3}$$

$$\Rightarrow 6xy + x^2 + 2 = 0$$

2. **Ans (D)**

If $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, P be any point on the hyperbola. The tangent at P intersects the asymptote at A and B where C the centre of the hyperbola $\Rightarrow$ Area of $\Delta CAB = ab$
 $\therefore \text{Area} = (4\sqrt{2}) \times (3\sqrt{2}) = 24$ sq unit.

3. **Ans (B)**

$$\sim (\sim p \wedge q) \wedge (p \vee q)$$

$$= (p \vee \sim q) \wedge p \vee q$$

$$= p \vee (\sim q \wedge q)$$

$$= p \vee c \text{ (c contradiction)}$$

$$= p$$

4. **Ans (B)**

$$\text{Use result } PS_1 + PS_2 = PF_1 + PS_2 = 2a$$

5. **Ans (D)**

$$\left. \begin{aligned} a - bx + cx &= 0 \\ ay + b - cy &= 0 \\ az - bz - c &= 0 \end{aligned} \right\} \text{ has non trivial solution}$$

$$\begin{vmatrix} 1 & -x & x \\ y & 1 & -y \\ z & -z & -1 \end{vmatrix} = 0$$

$$-1 - yz + x(-y + yz) + x(-yz - z) = 0$$

$$\Rightarrow -1 - yz - xy - xz = 0$$

$$\Rightarrow xy + yz + zx = -1$$

6. **Ans (C)**

$$\text{Use result } \frac{d^2x}{dy^2} = \frac{-\frac{d^2y}{dx^2}}{\left(\frac{dy}{dx}\right)^3}$$

7. **Ans (B)**

$$\text{Let } Q(x, y) \text{ on the curve } y^2 = 2x^3 + 9 - 3x^2$$

$$PQ^2 = (x - 1)^2 + y^2$$

$$PQ^2 = x^2 - 2x + 1 + 2x^3 + 9 - 3x^2$$

$$= 2x^3 - 2x^2 - 2x + 10$$

$$\frac{d}{dx} (PQ^2) = 6x^2 - 4x - 2 = 0$$

$$= (x - 1)(3x + 1) = 0$$

$$\Rightarrow x = 1, x = \frac{-1}{3}$$

$$\frac{d^2}{dx^2} ((PQ)^2) = 12x - 4$$

$$\text{at } x = 1 \text{ } PQ^2 \text{ is minimum}$$

$$PQ^2 = 8 \Rightarrow PQ = 2\sqrt{2}$$

8. **Ans (C)**

$$\text{Let } \vec{p} = a\hat{i} + b\hat{j} + c\hat{k}$$

$$\vec{q} = x\hat{i} + y\hat{j} + z\hat{k}$$

Use Inequality

$$|\vec{p} \times \vec{q}| = |\vec{p}| |\vec{q}| \sin \theta$$

$$\left| \vec{p} \times \vec{q} \right| \leq \left| \vec{p} \right| \left| \vec{q} \right|$$

$$|\vec{p} \times \vec{q}|^2 = (az - cx)^2 + (ay - bx)^2 + (bz - cy)^2$$

$$\leq (a^2 + b^2 + c^2)(x^2 + y^2 + z^2)$$

9. **Ans (B)**

$$\text{Here } f(g(x)) = \begin{cases} 1 - \sqrt{x}, & x \text{ is rational} \\ (1-x)^2, & x \text{ is irrational} \end{cases}$$

$(1-x)^2$ is always positive

$$\text{Let } f(g(x)) = -\sqrt{2}$$

$$1 - \sqrt{x} = -\sqrt{2}$$

$$x = 3 + 2\sqrt{2}$$

which is irrational and we take x is rational so

this is not possible so $f(g(x)) \neq -\sqrt{2}$

$f(g(x))$ is into function

$$\text{at } x = \frac{1}{4}$$

$$f\left(g\left(\frac{1}{4}\right)\right) = \frac{1}{2}$$

$$\text{at } x = 1 + \frac{1}{\sqrt{2}}$$

$$f\left(g\left(1 + \frac{1}{\sqrt{2}}\right)\right) = \frac{1}{2}$$

$f(g(x))$ is many one function

10. **Ans (B)**

$$f(x) > 0 \quad \forall x \in \mathbb{R} \quad \text{Put } x \rightarrow 2x \text{ \& } y \rightarrow 0$$

$$f(x) = \sqrt{f(2x)}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f\left(\frac{2x+2h}{2}\right) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{f(2x)f(2h)} - f(x)}{h}$$

$$= f(x) \lim_{h \rightarrow 0} \left(\frac{\sqrt{f(2h)} - 1}{h} \right)$$

$$= f(x) \lim_{h \rightarrow 0} \frac{f(2h) - 1}{2h} \times \frac{2}{(\sqrt{f(2h)} + 1)}$$

$$f'(x) = f(x) \cdot f'(0)$$

$$f'(x) = 2f(x)$$

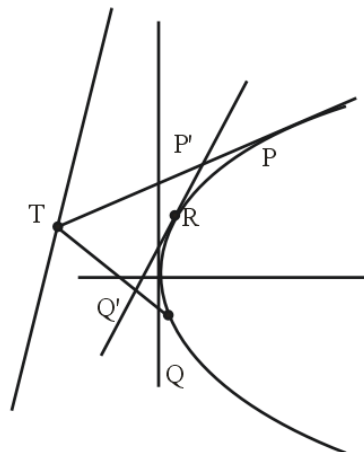
$$f(x) = e^{2x+c} \quad f(0) = 1 \Rightarrow c = 0$$

$$f(x) = e^{2x}$$

11. **Ans (D)**

$$\text{Use Property } \frac{TP'}{TP} + \frac{TQ'}{TQ} = 1$$

use A.M $\geq$ G.M.



12. **Ans (B)**

$$\text{Here } f(x) = 8x^3 + 4x^2 + x + 1$$

$$f'(x) = 24x^2 + 8x + 1 > 0 \quad \forall x \in \mathbb{R}$$

So the coefficient of highest power of x in $f^{-1}(x)$

$$\text{is } \frac{1}{2}$$

13. **Ans (C)**

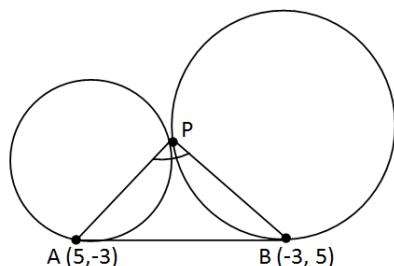
Now

$$\begin{aligned} \frac{B_n - B_{n-1}}{A_{n-1}} + \frac{1}{n} \frac{A_n}{B_n} &= \frac{\frac{\cos n\theta}{\cos^n \theta} - \frac{\cos(n-1)\theta}{\cos^{n-1} \theta}}{\frac{\sin(n-1)\theta}{\cos^{n-1} \theta}} + \frac{1}{n} \tan(n\theta) \\ &= \frac{2 \cos n\theta - 2 \cos(n-1)\theta \cdot \cos \theta}{2 \sin(n-1)\theta \cdot \cos \theta} + \frac{1}{n} \tan n\theta \\ &= \frac{2 \cos n\theta - \cos n\theta - \cos(n-2)\theta}{2 \sin(n-1)\theta \cdot \cos \theta} + \frac{1}{n} \tan n\theta \\ &= \frac{\cos n\theta - \cos(n-2)\theta}{2 \sin(n-1)\theta \cdot \cos \theta} + \frac{1}{n} \tan n\theta \\ &= \frac{-2 \sin(n-1)\theta \cdot \sin \theta}{2 \sin(n-1)\theta \cdot \cos \theta} + \frac{1}{n} \tan n\theta \\ &= -\tan \theta + \frac{\tan n\theta}{n} \end{aligned}$$

14. **Ans (C)**

$\angle APB$ is a right angle

$\Rightarrow$ locus of P is a circle whose end points of diameter are A and B



15. **Ans (B)**

$$P = \frac{10C_3 + 10C_2 \times 10C_1}{20C_3}$$

$$= \frac{1}{2}$$

16. **Ans (B)**

$\angle PBQ = \angle PAQ = \theta$

$\Rightarrow$ PBAQ is Cyclic

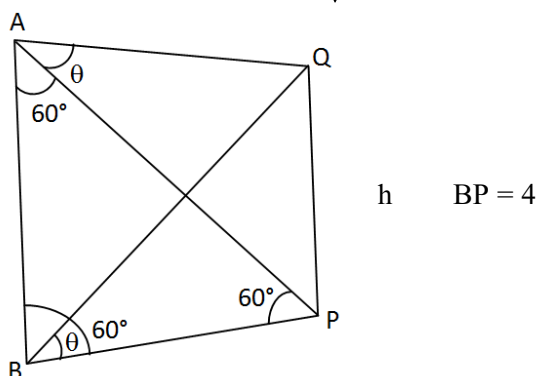
$\angle BAQ = 180^\circ - \angle BPQ = 90^\circ$

$\angle BQP = \angle BAP = 60^\circ$

$\theta = 90^\circ - \angle BQP = 30^\circ$

$$\tan \theta = \frac{h}{BP} = \frac{h}{4}$$

$$\Rightarrow h = 4 \tan \theta = 4 \times \frac{1}{\sqrt{3}} = \frac{4\sqrt{3}}{3} \text{ m}$$



17. **Ans (B)**

$A \times A = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (2, 3), (3, 2), \dots\}$

No. of antisymmetric relation which contain (2, 3) is $2^5 - 1 = 31$

18. **Ans (B)**

$$\int_0^{\pi/2} \frac{4 \cos^4 \theta}{\cos^4 \theta + \sin^2 2\theta} d\theta$$

$$= 4 \int_0^{\pi/2} \frac{\cos^2 \theta}{\cos^2 \theta + 4 \sin^2 \theta} d\theta$$

$$= 4 \int_0^{\pi/2} \frac{\sec^2 \theta}{(1 + \tan^2 \theta)(1 + 4 \tan^2 \theta)} d\theta \quad \tan \theta = x$$

$$= \int_0^\infty \frac{1}{(1 + x^2)(1 + 4x^2)} dx$$

$$= \int_0^\infty \left(\frac{-\frac{1}{3}}{1 + x^2} + \frac{\frac{1}{3}}{1 + 4x^2} \right) dx$$

$$= 4 \times \frac{\pi}{6} = \frac{2\pi}{3}$$

19. **Ans (B)**

$$= \int e^{x^4} (1 + x^2 + 2x^4) e^{x^2} 2x dx$$

$$= 2 \int e^{x^4 + x^2} (x + x^3 + 2x^5) dx$$

$$= 2 \int e^{x^4 + x^2} x dx + \int x^2 \cdot e^{x^4 + x^2} \times (2x + 4x^3) dx$$

$$= 2 \int e^{x^4 + x^2} x dx + x^2 e^{x^4 + x^2} - \int 2x e^{x^4 + x^2} dx$$

$$= x^2 e^{x^4 + x^2} + c$$

Here

$$f(x) = x^2 e^{x^4 + x^2}$$

$$f(1) + f'(0) = e^2$$

20. **Ans (C)**

For $x \in [0, 1]$

$$\cos^{-1} x + \sin^{-1} x + \tan^{-1} x = \frac{\pi}{2} + \tan^{-1} x$$

true

for $x \in [-1, 0)$

$$\cos^{-1} x - \sin^{-1} x - \tan^{-1} x = \frac{\pi}{2} + \tan^{-1} x$$

$$\text{L.H.S} > \frac{\pi}{2}, \quad \text{R.H.S} < \frac{\pi}{2} \text{ not possible}$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (171)**

$$\text{Here } n_1 = 35 \quad \bar{x}_1 = 80 \quad \sigma_1 = 4$$

$$n_2 = 65 \quad \bar{x}_2 = 70 \quad \sigma_2 = 3$$

$$\text{Combined } \bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2} = 73.5$$

$$\sigma^2 = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2},$$

$$d_1 = \bar{x}_1 - \bar{x}$$

$$\begin{aligned} d_2 &= \bar{x}_2 - \bar{x} \\ &= \frac{35(16 + 42.25) + 65(9 + 12.25)}{100} \end{aligned}$$

$$= 34.20$$

$$\sigma = \sqrt{34.20}$$

2. **Ans (728)**

$$AB = BA^3$$

$$(AB)B = (BA^3)B$$

$$= BA^2(AB)$$

$$AB^2 = BA^2 BA^3$$

$$= BA(AB) A^3$$

$$= BA BA^3 A^3$$

$$= B (BA^3)A^6$$

$$AB^2 = B^2 A^9$$

$$\text{In general } AB^n = B^n A^{3^n}$$

$$\text{put } n = 6,$$

$$AB^6 = B^6 A^{3^6}$$

$$A = A^{3^6} = A^{729}$$

$$A^{728} = I$$

$$K = 728$$

3. **Ans (15)**

$$xyz(x + 2y + 3z - 6) = 0$$

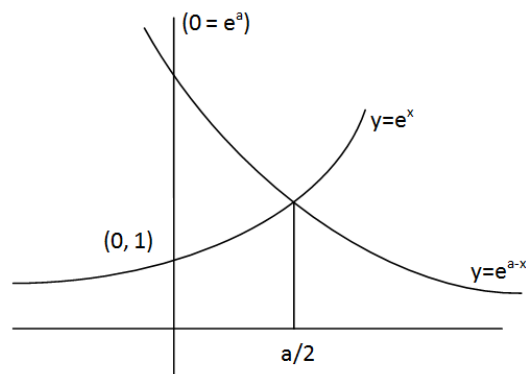
$$x = 0, y = 0, z = 0 \quad x + 2y + 3z = 6$$

$$x = 0, z = 0 \text{ and } y = 0 \text{ form 8 - octants and}$$

$x + 2y + 3z = 6$ divides each octants in two portions except one.

$$\text{Total no. of portion} = 2 \times 7 + 1 = 15$$

4. **Ans (3)**



$$\begin{aligned} S &= \int_0^{a/2} (e^{a-x} - e^x) dx \\ &= (-e^{a-x} - e^x)_0^{a/2} \\ &= e^a - 2e^{a/2} + 1 = (e^{a/2} - 1)^2 \end{aligned}$$

$$\lim_{a \rightarrow 0} \frac{S}{a^2} = \frac{1}{4}$$

5. **Ans (1)**

$$\text{put } x = 0, y = 1 \Rightarrow \text{get } f(1) = 0$$

$$\begin{aligned} f'(x) &= \lim_{n \rightarrow 0} \frac{f(x+n) - f(x)}{n} \\ &= \lim_{n \rightarrow 0} \frac{f(xn+1) - f(x) \cdot f(n) - f(x)}{n} \\ f(0) &= -1 \end{aligned}$$

$$f(1) = 0$$

$$\begin{aligned} &= \lim_{n \rightarrow 0} \frac{f(xn+1) - f(1)}{n} - \lim_{n \rightarrow 0} f(x) \frac{(f(n)+1)}{n} \\ &= xf'(1) - f'(0) \cdot f(x) \end{aligned}$$

$$f'(x) = x - f(x)$$

$$f'(x) + f(x) = x$$

$$f(x) e^x = x e^x - e^x + c$$

$$f(x) = x - 1 + e^{-x} c, \quad f(1) = 0 \Rightarrow c = 0$$

$$f(x) = x - 1$$

6. Ans (1)

$$f'(x) = |x| - \{x\}$$

$$= \begin{cases} x - \{x\} & , \quad x \geq 1 \\ 0 & , \quad 0 \leq x < 1 \\ -1 - 2x & , \quad -1 < x < 0 \\ -x - \{x\} & , \quad x \leq -1 \end{cases}$$

$$-1 - 2x < 0 \text{ for } -\frac{1}{2} < x < 0$$

$$f'(x) < 0 \text{ for } x \in \left(-\frac{1}{2}, 0\right)$$

7. Ans (2)

$$\text{Let } N = 32^{128} = (2^5)^{128} = 2^{640}$$

$$= 4^{320} = 3m + 1$$

$$(16)^{3m+1} = 2^{12m+4} = 2 \cdot 2^{12m+3} = 2(8^{4m+1}) = 2(7p+1)$$

Remainder is 2

8. Ans (1)

$$\frac{3}{4} + \frac{1}{6} + \frac{5}{96} + \frac{6}{320} + \dots$$

$$= \frac{3}{1 \times 2} \times \frac{1}{2} + \frac{4}{2 \times 3} \times \left(\frac{1}{2}\right)^2 + \frac{5}{3 \times 4} \times \left(\frac{1}{2}\right)^3 + \frac{6}{4 \times 5} \times \left(\frac{1}{2}\right)^4 + \dots$$

$$= \sum_{k=1}^n \frac{K+2}{K(K+1)} \left(\frac{1}{2}\right)^k = \sum_{k=1}^n \frac{2(K+1) - K}{K(K+1)} \left(\frac{1}{2}\right)^k$$

$$= \sum_{k=1}^n \left(\frac{1}{K} \left(\frac{1}{2}\right)^{k-1} - \frac{1}{(k+1)} \left(\frac{1}{2}\right)^k \right)$$

$$= 1 - \frac{1}{2} \left(\frac{1}{2}\right)^1 + \frac{1}{2} \left(\frac{1}{2}\right)^1 - \frac{1}{3} \left(\frac{1}{2}\right)^2 + \dots + \frac{1}{n} \left(\frac{1}{2}\right)^{n-1} - \frac{1}{n+1} \left(\frac{1}{2}\right)^n$$

$$= 1 - \frac{1}{n+1} \left(\frac{1}{2}\right)^n$$

$$= \lim_{n \rightarrow \infty} 1 - \frac{1}{n+1} \left(\frac{1}{2}\right)^n = 1$$

9. Ans (2)

The mirror image of A about the angle bisector

$y = x$ and $y = -2x$ are P and Q lies on side BC

$$P(2,1) \quad Q\left(\frac{-11}{5}, \frac{2}{5}\right)$$

Eqⁿ of BC is $x - 7y + 5 = 0$ angle bisector of B

and C meet at I(0,0), r = perpendicular distance

from I(0,0) to BC.

$$r = ID = \left| \frac{5}{\sqrt{1+49}} \right| = \frac{5}{5\sqrt{2}} = \frac{1}{\sqrt{2}}$$

10. Ans (25)

$$\text{let } \begin{cases} a = 2k_1 + 1 \\ b = 2k_2 + 1 \\ c = 2k_3 + 1 \\ d = 2k_4 + 1 \end{cases} \quad k_1, k_2, k_3, k_4 \text{ are non}$$

negative integer

$$16 \leq 2k_1 + 1 + 2k_2 + 1 + 2k_3 + 1 + 2k_4 + 1 \leq 36$$

$$6 \leq k_1 + k_2 + k_3 + k_4 \leq 16$$

no. of solutions

$$= 9_{c_3} + 10_{c_3} + \dots + 19_{c_3}$$

$$= 20_{c_4} - 9_{c_4}$$

$$n = 20, k = 9, m = 4$$

$$20 + 9 - 4 = 29 - 4$$

$$= 25$$